

Backtesting a strategy that combines **Ichimoku Cloud + Keltner Channel + True Strength Index (TSI)** is a good way to see whether the rules actually have an edge before trading with real money.

Step 1: Define Your Trading Rules

You need precise rules so every trade is evaluated consistently.

Buy Entry

Enter a **Call Buy** only when all of these conditions are true:

1. Ichimoku Cloud

- Price closes above the cloud.
- Tenkan-sen crosses above Kijun-sen.
- Chikou Span is above the price from 26 periods ago.
- Senkou Span A is above Senkou Span B (bullish cloud).

2. Keltner Channel

- Price closes above the middle EMA.
- Candle closes above the upper Keltner Channel.

3. True Strength Index (TSI)

- TSI crosses above its signal line.
- TSI is above zero.

4. Volume (Recommended)

- Current volume is higher than the 20-period average.
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Sell / Put Entry

Reverse all the conditions:

- Price below cloud.
 - Tenkan below Kijun.
 - Chikou below price.
 - Bearish cloud.
 - Candle closes below lower Keltner Channel.
 - TSI crosses below signal line.
 - TSI below zero.
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Step 2: Define Exit Rules

Example:

- Stop Loss = 5%
 - Target = 20%
 - Or exit when:
 - TSI gives opposite crossover.
 - Price closes inside the cloud.
 - Trailing stop is hit.
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Step 3: Choose Historical Data

For NIFTY options, you need:

- 1-minute candles (intraday)
- 5-minute candles (recommended)
- 15-minute candles

Test at least:

- 6–12 months
- Bullish market
- Bearish market
- Sideways market

Step 4: Backtest Manually

Use **TradingView Bar Replay**.

For every trade, record:

Date	Time	Strike	Buy Price	Sell Price	Profit	Win/Loss
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Continue until you have **at least 100 trades**.

Step 5: Calculate Results

Track these metrics:

- Total Trades
- Winning Trades
- Losing Trades
- Win Rate
- Average Profit
- Average Loss
- Maximum Drawdown
- Profit Factor
- Net Profit

Example

Metric	Value
Trades	150
Wins	93
Losses	57
Win Rate	62%
Profit Factor	1.85
Max Drawdown	8%

Step 6: Improve the Strategy

Test different settings:

Ichimoku

- Default: 9, 26, 52
- Also try:
 - 7, 22, 44
 - 10, 30, 60

Keltner Channel

- EMA Length: 20
- ATR Length: 10
- Multiplier:
 - 1.5
 - 2.0
 - 2.5

TSI

- Long = 25
- Short = 13
- Signal = 13

Also test:

- 20 / 10 / 10
 - 30 / 15 / 15
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Step 7: Avoid Common Mistakes

- Don't look ahead when replaying charts.
 - Use realistic entry prices.
 - Include brokerage and taxes.
 - Include slippage.
 - Avoid changing rules after seeing the outcome of a trade.
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Step 8: Automate the Backtest

You can automate this in **Python** using:

- Angel One SmartAPI (or your broker's API)
- pandas
- pandas-ta (or TA-Lib)
- NumPy
- Backtesting.py (or Backtrader)

The workflow is:

1. Download historical OHLCV data.
2. Calculate Ichimoku, Keltner Channel, and TSI.
3. Check each candle against your entry/exit rules.
4. Simulate trades with stop-loss and target.

5. Generate a report with:

- Net Profit
- Win Rate
- Profit Factor
- Maximum Drawdown
- Sharpe Ratio
- Equity Curve
- Trade Log

Based on what you've shared previously

You mentioned that your option-buying strategy is:

- **NIFTY 50 options**
- **Call and Put buying**
- Premium between ₹200–₹300
- **Target: 20%**
- **Stop Loss: 5%**
- Data from **Angel One API**

A Python backtester can be built specifically around those rules, including filtering option premiums in the ₹200–₹300 range and testing both call and put trades.

If you're planning to automate trading later, it's usually best to build the **backtester first**, verify it on at least one year of historical data, and then use the same logic for live trading.